

Project Executable Files

Date	15-06-2026
Team ID	xxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine Using Machine Learning
Maximum Marks	2 Marks

Project Executable Files

This section contains all the executable and deployment files required for the Smart Agricultural Production Optimization Engine (OptiCrop) project. The project has been developed using Python, Machine Learning, and Flask. All necessary source code, trained model files, datasets, dependencies, and application files are organized properly so that the evaluator can execute the project successfully.

Step 1: Submission Checklist

S.N	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt (dependency file)	Yes
4	Dataset (Crop Recommendation Dataset)	Yes
5	Environment configuration file (.env.example if applicable)	NA
6	Deployed application URL (if hosted)	No

7 8	Executable application (Flask Web Application)	Yes
9	Dockerfile / Containerization configuration	NA
10	Test files and prediction results	Yes
	Demo video / Walkthrough	Yes

Step 2: File / Folder Structure

The Smart Agricultural Production Optimization Engine (OptiCrop) crop project follows a modular folder structure to separate datasets, machine learning models, frontend resources, backend logic, and supporting files. This organization improves maintainability, readability, and scalability while allowing each component to perform a specific role within the application.

```
OptiCrop/ |
├── dataset/
│   └── crop_recommendation.csv
├── models/
│   └── crop_model.pkl
├── static/
│   ├── css/
│   ├── js/
│   └── images/
├── templates/
│   └── index.html
├── app.py
├── model_training.py
├── preprocessing.py
├── requirements.txt
├── README.md
└── recommendation.py
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Deployed
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Flask Server
Repository Link	(Add your GitHub Repository URL)
Demo Video Link	(Add your Demo Video Link)

Step 4: Run Instructions

Follow the steps below to execute the project locally:

1. Download or clone the project repository.
2. Open the project folder in Visual Studio Code or PyCharm.
3. Create and activate a Python virtual environment (optional).
4. Install all required dependencies using:
`pip install -r requirements.txt`
5. Ensure the trained model file and dataset are available in the project directory.
6. Start the Flask application using:
`python app.py`
7. Open the browser and visit:
`http://127.0.0.1:5000`
8. Enter the soil and environmental parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH level, and rainfall.
9. Click the Predict button to view the recommended crop.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Crop recommendation accuracy depends on the quality and size of the agricultural dataset.	Use a larger and updated agricultural dataset.
2	The application supports local deployment only.	Can be deployed on cloud platforms such as Render or Heroku.
3	The system recommends crops based only on the provided input parameters and does not consider real-time weather or market conditions.	Future versions can integrate live weather APIs and market price data for better recommendations.